Value Creation as an Engineering Habit

In ECE 350, one of our goals is to help you build *engineering habits of mind* that extend beyond technical skills. One of the most important of these habits is **value creation**, a core element of the *Entrepreneurial Mindset* described in the Engineering Unleashed / KEEN framework:

https://engineeringunleashed.com/framework

Value creation is also a critical lens for answering behavioral interview questions. Employers are not only interested in *what* you did, but *why it mattered*, *to whom*, and *what changed because of your actions*. This document introduces what value is, the different kinds of value engineers create, and how to recognize value in your own experiences so you can communicate it effectively.

What Is "Value"?

Value is the benefit created for a stakeholder as a result of an action, decision, or solution. In engineering, value is not limited to money or profit. Engineers create value whenever their work improves outcomes, reduces risk, saves time or resources, increases understanding, or enables others to succeed.

A simple way to think about value is:

Value exists when something is better for someone because of what you did.

If nothing meaningfully changed for anyone, then value was likely not created, even if you completed a task or met a requirement.

Value Creation in the Engineering Unleashed / Entrepreneurial Mindset

Within the Entrepreneurial Mindset, **value creation** is one of the "3 Cs":

- **Curiosity** – exploring and questioning assumptions
- **Connections** – integrating information, people, and ideas
- **Creating Value** – producing solutions that benefit stakeholders

Creating value means asking:

- Who are the stakeholders?
- What needs, problems, or constraints do they have?
- How does my work improve their situation?

This mindset applies equally to startups, large companies, research labs, class projects, internships, and student organizations.

Types of Value Engineers Create

When reflecting on your experiences, consider *what kind of value* your contributions created. One project may create several types of value at once.

1. Technical / Functional Value

This is the most familiar form of value for engineers.

Examples:

- Improving system performance (speed, accuracy, efficiency)
- Fixing a bug or failure mode
- Designing a component that meets specifications

2. Economic / Resource Value

Value created by saving or using resources more effectively.

Examples:

- Reducing costs or material usage
- Saving time or labor hours
- Improving energy efficiency
- Preventing rework or delays

3. Risk Reduction / Reliability Value

Engineers often create value by preventing problems before they occur.

Examples:

- Increasing safety or reliability
- Identifying potential failures
- Adding validation, testing, or safeguards
- Improving robustness or fault tolerance

4. Learning and Knowledge Value

Sometimes the primary value is increased understanding.

Examples:

- Developing documentation or tutorials
- Transferring knowledge to teammates
- Prototyping to explore feasibility
- Clarifying requirements or assumptions

5. Team and Process Value

Not all value is technical. Engineers also create value through how work gets done.

Examples:

- Improving communication or coordination
- Organizing tasks or workflows
- Mentoring teammates
- Resolving conflict or ambiguity

6. User or Stakeholder Value

Value experienced directly by users, clients, or customers.

Examples:

- Improving usability or accessibility
- Responding to user feedback
- Adapting designs to real-world constraints
- Delivering something that met a real need

Engineers are not valuable because they complete tasks. Engineers are valuable because they create meaningful impact.

Learning to recognize and articulate value is a habit that will serve you in interviews, on the job, and throughout your career!!